Netflix Content Catalog Analysis

Data cleaning and EDA of 8,807 Netflix titles added up to September 2021 (Kaggle data)
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Executive summary

This project cleans and explores Netflix's catalog to understand how Netflix builds its content over time, by audience, by genre and by region, and what that suggests for its content strategy.

- **The catalog was built almost entirely from 2016 onwards** (98.4% of titles), peaking at 2,016 titles in 2019. The 2021 drop is a data artefact, not a real decline.
- **Content is aimed at mature audiences:** TV-MA (36%) and TV-14 (25%) lead; content for young children is only 6.5%.
- **The US leads, but Asia is rising:** the US has 46.2% of titles with country data; India ranks 2nd (13.1%), and the top 10 includes India, Japan and South Korea.
- **Movies dominate** (69.7% of titles), and about two thirds of TV Shows have only one season.

Data and approach

Item	Details
Data	netflix_titles.csv (Kaggle): 8,807 titles × 12 columns, titles added up to September 2021
Cleaning	Filled missing director, cast and country with "Unknown"; dropped 17 rows missing date_added, rating or duration (including 3 rows where duration was entered in the rating column); checked duplicates; parsed dates and split duration into number and unit; checked date_added vs release_year; reviewed outliers (kept as real values); grouped 14 rating codes into Kids / Teens / Adults
Questions	8 business questions on content type, duration, rating, growth by year, seasonality, directors, countries and description keywords
Tools	Python (Pandas, NumPy, Matplotlib, Seaborn, WordCloud), Jupyter

Key findings

1. The catalog was built from 2016 onwards

2016–2021 accounts for 98.4% of all titles, peaking at 2,016 titles in 2019. New titles appear to fall in 2021, but the data stops in September: the monthly pace in 2021 (about 166 titles per month) was close to the 2019 peak (168). The drop comes from incomplete data, not a change in strategy.

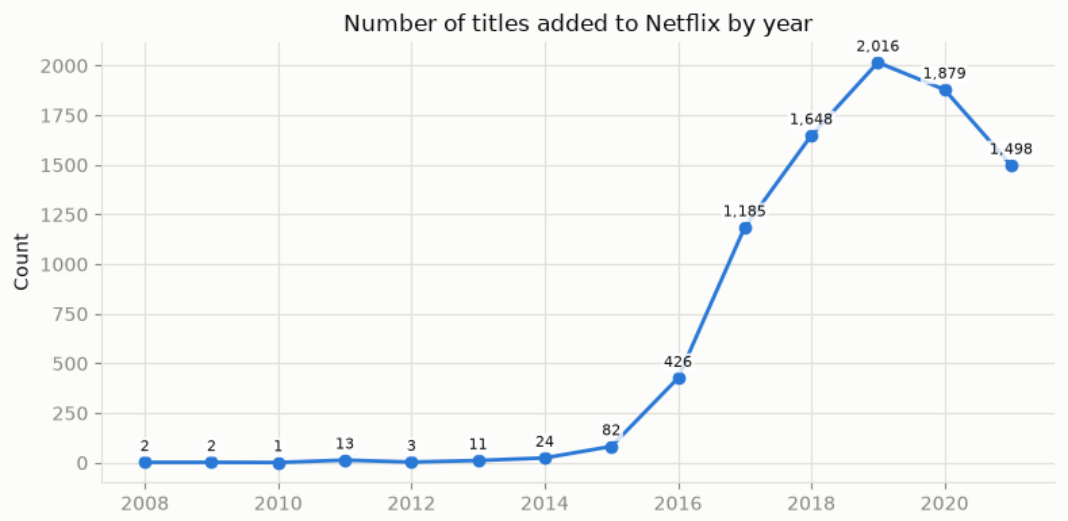


Figure 1. Number of titles added by year

2. Content is aimed at mature audiences

TV-MA (36%) and TV-14 (25%) are the most common ratings. Adults content makes up 45.6% of the catalog, while content for young children (TV-Y, TV-G, G) is only 6.5%.

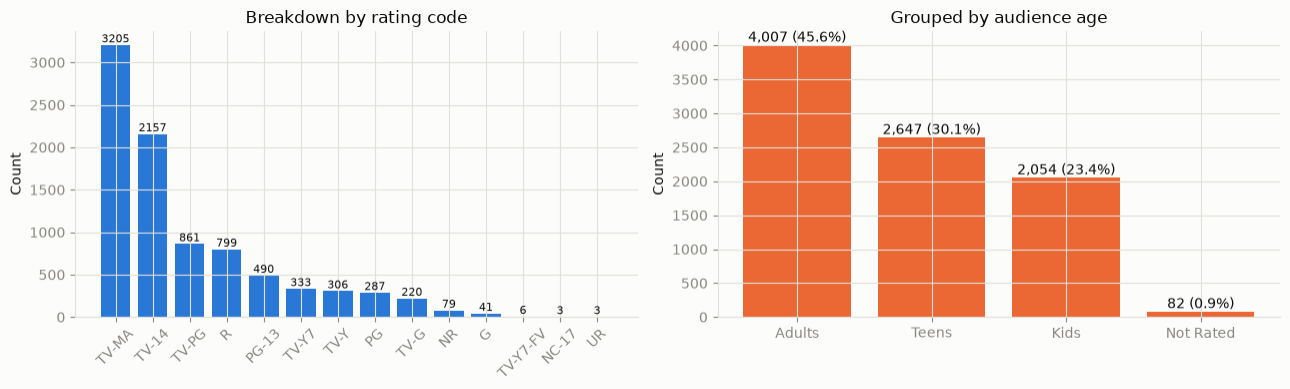


Figure 2. Rating breakdown

3. The US leads, but India and Asia are rising

The US accounts for 46.2% of titles with country information, 3.5 times the runner-up. India ranks 2nd (13.1%), ahead of the UK, and the top 10 includes three Asian countries (India, Japan, South Korea).

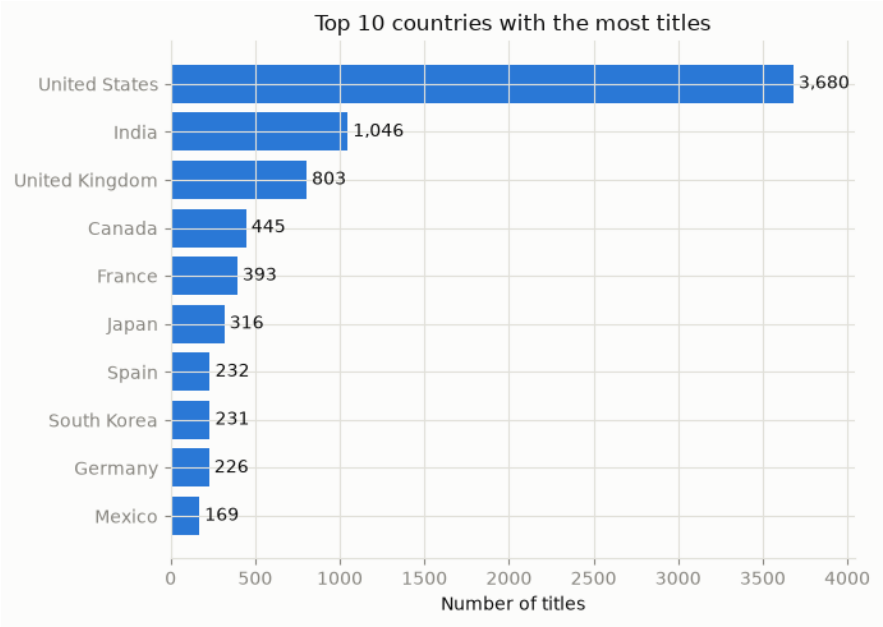


Figure 3. Top 10 countries by number of titles

4. Other findings

- Movies make up 69.7% of titles (6,126), about 2.3 times the number of TV Shows; about two thirds of TV Shows have only one season.
- No clear seasonality: each genre's monthly share stays within about 5–11% around the even 8.3% baseline.
- Top directors reflect format, not popularity: most make stand-up comedy specials or Indian children's animation, short formats that are easy to produce in volume.
- Descriptions sell people and emotions: the most common keywords revolve around family, love, youth and a character's journey.

Recommendations

- Consider expanding content for young children (only 6.5% of the catalog), after checking it against the share of accounts with kids' profiles.
- Keep investing in Asian content, and analyse year-by-year growth for India, Japan and South Korea to confirm the trend.

- Review the series renewal strategy with viewership data, to tell intentional short series from cancellations.
- Use low-cost formats such as stand-up comedy and short animation to grow the catalog, if viewer performance justifies it.
- Plan releases around marketing campaigns rather than seasons, since genres show no clear seasonality.

Limitations

- Data runs only to September 2021, so 2021 figures do not cover the full year.
- Movies are measured in minutes and TV Shows in seasons, so their durations cannot be compared directly.
- About 30% of rows are missing director and 9% are missing country, which may slightly skew those rankings.
- No viewership data, so the recommendations still need to be checked against audience performance.